

PRODUCT RISK ANALYSIS: Designing Robust Products

Those who are following the author's series of articles would have seen design of the charge controller in the previous article. In that article the design of the charge controller is a fully functional one, but it lacks safety features. A good product should not only function properly but must be robust. A well designed product will accept a certain amount of abuse and will continue to perform nicely. Such fault tolerant products are loved by the customers



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Any product will face variations from the design conditions during manufacture and use. Normally designers specify the tolerance limits of the design parameters, but a product design needs to be checked for proper performance with the worst case scenario of the process variations.

Apart from the designed tolerances, the product may be used in different manner than the designer had planned. A robust product will anticipate those unintended uses and provide features for such uses. The product should also have safety features to prevent damage or to minimise damage due to any unplanned usage.

We must appreciate that our effort

to make product robust comes at a cost. In most cases such a change will add to the cost, weight, and probably size of the product. Customers do not like to pay extra for things that are not directly delivering value. A good design becomes an art of compromise on cost versus quality.

We need to understand effects of various risks and prioritise our budget. We need to understand which failures are important and which can be ignored. Systematic evaluation of such risk and its effect is called failure mode effect analysis (FMEA).

FMEA

Several things can go wrong with a product. Our analysis begins with identi-